

Notes: Morris & Shin (AER, 1998)
Unique Equilibrium in a Model of Self-Fulfilling Currency Attacks

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1 Big picture

- **Question.** Why do currency attacks occur at some dates and not others when a currency has already been “ripe for attack” for a long time?
- **Problem with first-generation and standard self-fulfilling crisis models.** If fundamentals alone determine crises, the timing of attacks is hard to explain. If common-knowledge self-fulfilling beliefs determine crises, the model has multiple equilibria and cannot say which equilibrium will occur.
- **Main contribution.** Morris and Shin add a small amount of private noise to speculators’ information. This destroys common knowledge of fundamentals and selects a *unique* equilibrium.
- **Core mechanism.** A speculator cares not only about the fundamentals, but also about whether enough other speculators will attack. Noisy private signals make higher-order beliefs well-defined and generate a unique cutoff rule.
- **Equilibrium form.** There is a unique signal cutoff x^* : speculators attack if and only if their private signal $x < x^*$. There is also a unique fundamental cutoff θ^* : the government abandons the peg if and only if $\theta \leq \theta^*$.
- **Global-games insight.** Even arbitrarily small private noise can eliminate the continuum of self-fulfilling equilibria that arises under perfect common knowledge.
- **Policy value.** The unique equilibrium lets the authors analyze transaction costs, hot money, capital controls, and public announcements. These comparative statics are hard to study in a pure multiple-equilibrium model.
- **Main limiting result.** As signal noise tends to zero, the attack cutoff satisfies

$$f(\theta^*) = e^* - 2t,$$

where e^* is the peg, $f(\theta)$ is the floating exchange rate, and t is the cost of attacking.

- **Transparency implication.** What matters is not simply how much information speculators have, but whether information is publicly and commonly understood. A small “grain of doubt” about what others know can trigger an attack.
- **Economic takeaway.** Currency attacks can be self-fulfilling, yet still have a unique and analyzable equilibrium once speculators have private noisy information. The onset of a crisis is governed by higher-order beliefs, not just fundamentals or arbitrary shifts in sentiment.

2 Model objects and measurement

2.1 State of fundamentals

- The economy is summarized by a fundamental state $\theta \in [0, 1]$.
- Higher θ means stronger fundamentals.
- If the government abandons the peg, the floating exchange rate is $f(\theta)$.
- The function f is strictly increasing in θ .

Interpretation: A higher θ makes devaluation less profitable for speculators and makes defense easier for the government. The model compresses the macroeconomic state into one ordered index.

2.2 Pegged exchange rate and speculative payoff

- The currency is initially pegged at e^* .
- A speculator can either attack by short-selling one unit of the currency or refrain from attacking.
- Short-selling costs $t > 0$.
- If the peg is abandoned, an attacking speculator receives

$$e^* - f(\theta) - t.$$

- If the peg is defended, an attacking speculator receives $-t$.
- A nonattacking speculator receives zero.

Interpretation: Attacking is profitable only if the expected depreciation is large enough to cover the trading cost. This cost later becomes the policy variable behind Tobin-tax-style arguments.

2.3 Government payoff and defense cost

- The government obtains value $v > 0$ from maintaining the peg.
- It pays a cost $c(a, \theta)$ to defend the peg, where a is the proportion of speculators attacking.
- $c(a, \theta)$ is increasing in a : more selling pressure raises defense costs.
- $c(a, \theta)$ is decreasing in θ : stronger fundamentals make defense cheaper.
- The government defends if

$$v - c(a, \theta) \geq 0,$$

and abandons otherwise.

Interpretation: The government's problem creates strategic complementarity among speculators: if more traders attack, it becomes more likely the government abandons, which makes attacking more attractive.

2.4 Two benchmark thresholds

The paper defines two benchmark fundamentals.

- $\underline{\theta}$ solves

$$c(0, \underline{\theta}) = v.$$

If $\theta < \underline{\theta}$, the government abandons even with no attack.

- $\bar{\theta}$ solves

$$f(\bar{\theta}) = e^* - t.$$

If $\theta > \bar{\theta}$, even a successful attack does not cover the trading cost.

Interpretation: These thresholds divide fundamentals into unstable, stable, and strategically fragile regions. The middle region is where multiple equilibria arise under perfect information.

2.5 Tripartite classification of fundamentals

- **Unstable region** $[0, \underline{\theta}]$. The peg collapses regardless of speculator behavior.
- **Ripe-for-attack region** $(\underline{\theta}, \bar{\theta})$. If few speculators attack, the government defends. If many attack, the government abandons. This is the multiple-equilibrium region under common knowledge.
- **Stable region** $[\bar{\theta}, 1]$. Speculators do not attack because the devaluation payoff is too small to cover the cost t .

Interpretation: The paper's novelty is not the classification itself. Its contribution is showing how incomplete information selects a unique outcome inside the ripe-for-attack region.

2.6 Private signals and noise

- Nature chooses θ uniformly on $[0, 1]$.
- Each speculator observes a private signal

$$x_i \sim U[\theta - \varepsilon, \theta + \varepsilon],$$

independently across speculators conditional on θ .

- The noise $\varepsilon > 0$ is small.
- A speculator observes x_i but not θ or the signals of other speculators.

Interpretation: The small private noise is enough to remove common knowledge. Even if every trader's signal is very accurate, no one knows exactly what others saw, no one knows what others know about others' signals, and so on.

2.7 Aggregate attack strategy

- Let $\pi(x)$ be the proportion of speculators who attack after observing signal x .
- Given state θ , the aggregate fraction attacking is

$$s(\theta, \pi) = \frac{1}{2\varepsilon} \int_{\theta-\varepsilon}^{\theta+\varepsilon} \pi(x) dx.$$

- Let $a(\theta)$ be the critical mass of attackers that makes the government abandon the peg.
- The abandonment event is

$$A(\pi) = \{\theta : s(\theta, \pi) \geq a(\theta)\}.$$

Interpretation: The aggregate action is smooth in the fundamental because different speculators receive different noisy signals. This smoothing is crucial for the unique crossing argument.

3 Models and theoretical specifications

3.1 Payoff to attacking under an aggregate strategy

The state-contingent payoff from attacking is

$$h(\theta, \pi) = \begin{cases} e^* - f(\theta) - t, & \theta \in A(\pi), \\ -t, & \theta \notin A(\pi). \end{cases}$$

Given signal x , the expected payoff to attacking is

$$u(x, \pi) = \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} h(\theta, \pi) d\theta = \frac{1}{2\varepsilon} \int_{A(\pi) \cap [x-\varepsilon, x+\varepsilon]} (e^* - f(\theta)) d\theta - t.$$

Interpretation: A trader attacks when the expected depreciation, averaged over states consistent with her signal and conditional on the peg collapsing, exceeds the transaction cost.

3.2 Equilibrium condition for speculators

A strategy profile π is an equilibrium if

$$\pi(x) = 1 \quad \text{whenever } u(x, \pi) > 0,$$

$$\pi(x) = 0 \quad \text{whenever } u(x, \pi) \leq 0.$$

Interpretation: Equilibrium is a fixed point: a trader's best response depends on whether enough others attack, while the mass of attackers depends on individual best responses.

3.3 Lemma 1: strategic complementarity

If $\pi(x) \geq \pi'(x)$ for all x , then

$$u(x, \pi) \geq u(x, \pi') \quad \text{for all } x.$$

Interpretation: More aggressive behavior by other speculators enlarges the set of states in which the government abandons the peg. This raises the payoff from attacking for every signal. The game has strategic complementarities, but incomplete information prevents multiplicity.

3.4 Cutoff strategies

For any cutoff k , define the step strategy

$$I_k(x) = \begin{cases} 1, & x < k, \\ 0, & x \geq k. \end{cases}$$

Under I_k , aggregate selling is decreasing in θ :

$$s(\theta, I_k) = \begin{cases} 1, & \theta \leq k - \varepsilon, \\ \frac{1}{2} - \frac{\theta - k}{2\varepsilon}, & k - \varepsilon < \theta < k + \varepsilon, \\ 0, & \theta \geq k + \varepsilon. \end{cases}$$

Interpretation: Low fundamentals generate many low signals, so many speculators attack. High fundamentals generate high signals, so few attack. The aggregate attack curve slopes down.

3.5 Lemma 2: monotonicity of marginal payoff

The paper proves that

$$u(k, I_k)$$

is continuous and strictly decreasing in k .

Interpretation: A higher cutoff corresponds to a stronger marginal signal. When the marginal signal is stronger, the state is likely stronger, the peg is less likely to collapse, and the gain from short-selling is lower.

3.6 Lemma 3: unique signal cutoff

There is a unique x^* such that in every equilibrium,

$$\pi(x) = I_{x^*}(x).$$

Equivalently, a speculator attacks if and only if

$$x < x^*.$$

Interpretation: The proof uses Lemma 1 to squeeze any possible equilibrium strategy between two cutoff strategies and Lemma 2 to show that the only possible cutoff is the unique solution of $u(k, I_k) = 0$.

3.7 Theorem 1: unique fundamental cutoff

There is a unique θ^* such that the government abandons the peg if and only if

$$\theta \leq \theta^*.$$

Interpretation: The private-signal game still has self-fulfilling features, but it no longer has multiple equilibria. The equilibrium selects one critical state of fundamentals. Below it, attack succeeds; above it, attack fails.

3.8 Theorem 2: limiting cutoff as noise vanishes

As $\varepsilon \rightarrow 0$, the fundamental cutoff satisfies

$$f(\theta^*) = e^* - 2t.$$

Interpretation: The marginal speculator near θ^* assigns approximately probability one half to peg abandonment. Indifference requires half the depreciation gain to equal the attack cost t . Hence $\frac{1}{2}(e^* - f(\theta^*)) = t$.

3.9 Comparative statics: transaction costs

Differentiating the limiting condition gives

$$f'(\theta^*) \frac{d\theta^*}{dt} = -2, \quad \frac{d\theta^*}{dt} = -\frac{2}{f'(\theta^*)} < 0.$$

Interpretation: Higher transaction costs lower the attack cutoff. Therefore, fewer states lead to attacks. The effect is larger when $f'(\theta^*)$ is small, meaning the exchange-rate payoff is not very sensitive to fundamentals.

3.10 Comparative statics: aggregate wealth of speculators

An increase in hot money lowers the critical fraction $a(\theta)$ needed to force abandonment. This enlarges $A(\pi)$ and raises the payoff to attacking.

$$a(\theta) \downarrow \Rightarrow A(\pi) \uparrow \Rightarrow u(x, \pi) \uparrow \Rightarrow \theta^* \uparrow.$$

Interpretation: More speculative wealth makes attacks easier to coordinate and increases the range of fundamentals under which the peg collapses. However, the effect is most important when information noise is large.

4 Theoretical design and identification logic

4.1 What the paper identifies theoretically

- The paper identifies the equilibrium-selection force in a self-fulfilling crisis model.
- The force is not irrationality, sunspots, or exogenous shifts in sentiment.
- The force is the failure of common knowledge produced by tiny private information noise.
- The model maps primitive parameters into one unique crisis cutoff.

Interpretation: This is a theoretical identification exercise: the authors identify which ingredient makes the onset of crisis predictable within the model.

4.2 Why perfect information gives multiplicity

- In the unstable region, attack is unavoidable.
- In the stable region, attack is never profitable.
- In the ripe-for-attack region, two outcomes can be self-confirming under common knowledge.
- If everyone believes others will not attack, the government defends and no attack is optimal.
- If everyone believes others will attack, the government abandons and attack is optimal.

Interpretation: Multiplicity arises because the model assumes speculators commonly know the state and commonly know that others know it. The paper's central move is to relax this assumption only slightly.

4.3 Why private noise restores uniqueness

- A trader with a bad signal expects that many other traders also received bad signals.
- A trader with a good signal expects that few others received bad signals.
- The marginal trader is pinned down by indifference.
- Since the marginal payoff is strictly decreasing in the signal cutoff, the cutoff is unique.

Interpretation: Private noise converts a discontinuous coordination problem into a continuous threshold problem.

4.4 Role of higher-order beliefs

- Even if everyone knows fundamentals are not terrible, it may not be common knowledge that everyone knows this.
- If a signal is above $\underline{\theta} + \varepsilon$, the trader knows fundamentals are not in the unstable region.
- To know that everyone knows this, the signal must be higher, roughly $\underline{\theta} + 3\varepsilon$.
- To know that everyone knows that everyone knows this, the signal must be higher still.
- For any finite signal and positive ε , some sufficiently high order of knowledge fails.

Interpretation: This is the key global-game logic. Tiny private noise matters because coordination depends on beliefs about beliefs, not only first-order beliefs about fundamentals.

4.5 Policy interpretation

- Public announcements can matter because they restore common knowledge, not merely because they change average beliefs.
- Transaction taxes or trading costs can reduce attacks by reducing speculative payoff.
- Capital controls that raise the effective critical mass $a(\theta)$ are more effective when fundamentals are opaque and signal dispersion is large.
- When information is very precise, direct controls on hot money have less impact on the limiting cutoff.

Interpretation: The model changes the policy question from “which equilibrium will beliefs select?” to “how do policy instruments shift the unique cutoff?”

4.6 What the paper cannot fully prove

- The model is theoretical and does not estimate crisis probabilities empirically.
- Signals are assumed uniform and independent; real information networks may be more complicated.
- Speculators are modeled as a continuum, so individual strategic market power is absent.
- The government observes fundamentals and the attack mass perfectly, while real governments face their own information and credibility problems.
- The model is static; it abstracts from dynamic reserves, learning, policy reputation, and sequential attacks.

Interpretation: The paper’s strength is a clean mechanism for equilibrium selection, not a full empirical model of all currency crises.

5 Section-by-section study guide

5.1 Introduction

The introduction motivates the puzzle. Currency crises often happen suddenly, even when fundamentals looked vulnerable for a long period. Standard self-fulfilling models say attacks happen because speculators expect attacks, but this leaves the onset of the attack unexplained. Morris and Shin argue that noisy private information and higher-order uncertainty make the onset endogenous.

5.2 Model

The model defines the government, speculators, the peg, the floating exchange rate, the cost of defense, and the cost of attacking. It first classifies fundamentals under complete information. The middle region is the important one: the currency is viable if speculators stay away but collapses if enough speculators attack.

5.3 Game with imperfect information

Speculators receive noisy private signals. This generates uncertainty about other traders' signals and therefore about their behavior. The government abandons the peg if the realized mass of attackers exceeds the critical mass. The speculator's payoff becomes an expectation over both fundamentals and the likelihood of collective attack.

5.4 Unique equilibrium

The paper proves uniqueness through three steps: strategic complementarity, monotonicity of the marginal cutoff payoff, and the cutoff equilibrium. The unique equilibrium is a threshold strategy. This is the mathematical core of the paper.

5.5 Comparative statics and policy

The paper shows how the cutoff changes with information, transaction costs, and speculative wealth. Public signals and transparency are emphasized because common knowledge is crucial. Transaction costs reduce attacks. Hot money increases attacks, but its effect is muted when private signals are very precise.

5.6 Conclusion and appendix

The conclusion contrasts the model with fundamentals-only models and pure self-fulfilling-beliefs models. The appendix proves Lemma 2 and Theorem 2. The appendix is useful because it shows exactly why the marginal payoff is decreasing and why the limiting cutoff is $f(\theta^*) = c^* - 2t$.

6 Tables and figures

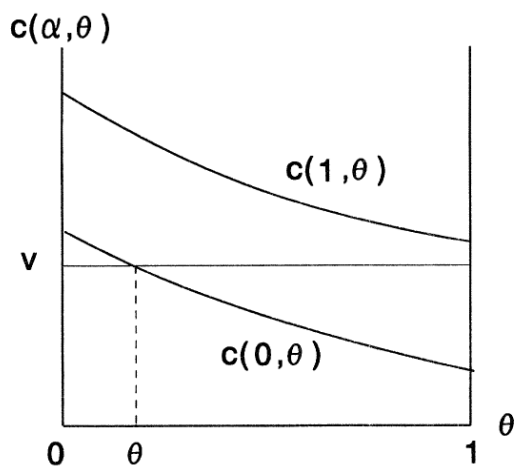
6.1 Figure 1: Cost and Benefit to the Government in Maintaining the Currency Peg + Figure 2: The Managed Exchange Rate and the Exchange Rate in the Absence of Intervention as a Function of the State of Fundamentals

Read:

- Figure 1 plots the government's defense cost $c(a, \theta)$ against the fundamental θ for different attack intensities.
- The horizontal value line v marks the benefit of maintaining the peg.
- The crossing of $c(0, \theta)$ and v defines the low-fundamental threshold $\underline{\theta}$.
- When $\theta < \underline{\theta}$, even zero attack pressure leaves defense too costly, so the peg is fundamentally unsustainable.
- Figure 2 plots the peg e^* , the speculative profitability line $e^* - t$, and the floating rate $f(\theta)$.
- The crossing of $f(\theta)$ with $e^* - t$ defines $\bar{\theta}$.
- When $\theta > \bar{\theta}$, a successful devaluation is too small to cover the cost of attacking.

Detailed interpretation: Figure 1 identifies the government's fundamental constraint. A weak state makes defending the peg costly even without short sales. More short sales shift the cost curve upward, so even stronger fundamentals can become vulnerable if enough speculators attack. Figure 2 identifies the speculator's payoff constraint. Even if speculators could force a devaluation, they do not want to attack when the floating rate is close to the peg because the trading cost wipes out the gain.

Economic message: Together, the two figures divide fundamentals into unstable, ripe-for-attack, and stable regions. The ripe-for-attack region is where the model's coordination problem lives.



(a) Figure 1: Cost and benefit to the government.

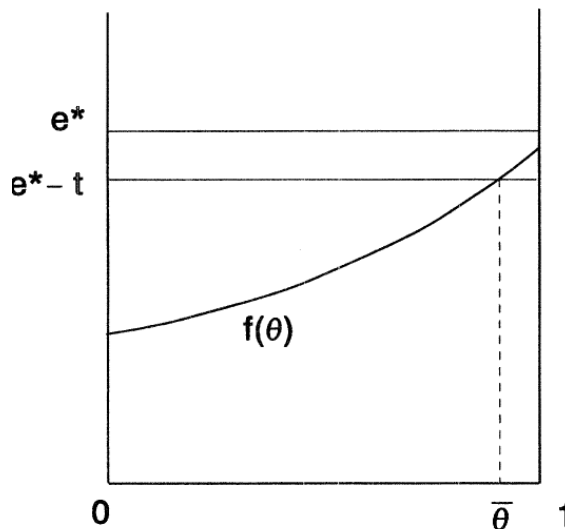


FIGURE 2. THE MANAGED EXCHANGE RATE AND THE

(b) Figure 2: Peg, floating rate, and attack profitability.

6.2 Figure 3: The Proportion of Speculators Whose Short Sales Are Sufficient to Induce Depreciation, Expressed as a Function of the Fundamentals

Read:

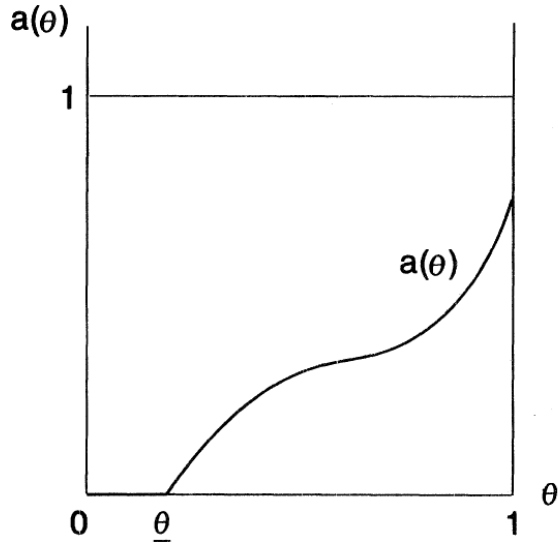
- Figure 3 plots $a(\theta)$, the critical fraction of speculators needed to force devaluation.
- In the unstable region, $a(\theta) = 0$: no attack is needed for abandonment.
- In the ripe-for-attack region, $a(\theta)$ rises with fundamentals.
- Stronger fundamentals require a larger attack mass to induce the government to abandon the peg.
- The function is bounded above by one; if even everyone attacking cannot force abandonment, the peg is effectively robust.

Detailed interpretation: This figure converts the government's optimization problem into a reduced-form attack threshold. It is the bridge from the government side of the model to the speculator game. Once $a(\theta)$ is known, speculators only need to reason about whether their collective attack mass $s(\theta, \pi)$ will exceed $a(\theta)$.

Economic message: The slope of $a(\theta)$ captures how fundamentals strengthen the government's ability to resist attack. Lowering $a(\theta)$ through more speculative wealth or more mobile capital raises crisis risk.

IRIUM AND CURRENCY ATTACKS

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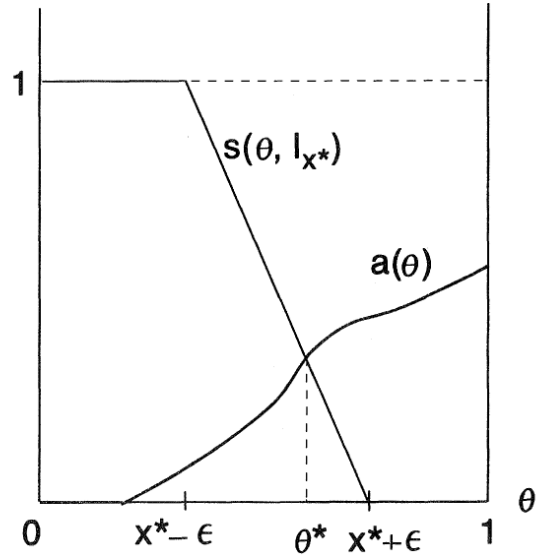


FIGURE 4. THE DERIVATION OF THE CUTOFF POINT FOR THE STATE OF FUNDAMENTALS AT WHICH THE

6.3 Figure 4: The Derivation of the Cutoff Point for the State of Fundamentals at Which the Equilibrium Short Sales Are Equal to the Short Sales Which Induce Depreciation

Read:

- Figure 4 plots two curves: equilibrium aggregate short sales $s(\theta, I_{x^*})$ and the critical attack mass $a(\theta)$.

- Aggregate short sales decline with θ because high fundamentals generate high signals, so fewer speculators attack.
- The critical mass $a(\theta)$ increases with θ because strong fundamentals make defense easier.
- The two curves cross once.
- The crossing defines the unique fundamental cutoff θ^* .
- The government abandons for $\theta \leq \theta^*$ and defends for $\theta > \theta^*$.

Detailed interpretation: This is the paper's most important diagram. It shows why the private-information game has a unique outcome. The downward-sloping attack mass and the upward-sloping resistance threshold can cross only once. That crossing pins down the unique crisis point. The diagram also explains comparative statics: shifting $a(\theta)$ down raises θ^* , while raising transaction costs lowers the signal cutoff and reduces attacks.

Economic message: The unique equilibrium is not imposed by assumption. It arises because noisy signals generate a monotone aggregate attack curve that intersects the government's resistance curve at one point.

7 Short summary

- The paper studies self-fulfilling currency attacks with speculators who have noisy private signals.
- Under complete information, the ripe-for-attack region has multiple equilibria.
- Small private noise eliminates common knowledge and selects a unique equilibrium.
- Speculators follow a cutoff strategy: attack if their signal is below x^* .
- The government follows a cutoff outcome: abandon the peg if fundamentals are below θ^* .
- The critical mass of attackers $a(\theta)$ rises with fundamentals.
- The actual attack mass $s(\theta, I_{x^*})$ falls with fundamentals.
- The unique crossing between these two curves determines the crisis cutoff.
- As signal noise vanishes, the cutoff satisfies $f(\theta^*) = e^* - 2t$.
- Higher transaction costs lower the cutoff and reduce the range of fundamentals in which attacks occur.
- More hot money lowers the critical attack mass and can raise the cutoff.
- Public announcements matter because they can restore common knowledge of fundamentals.
- The model explains why subtle news events can trigger crises even when average fundamentals appear sound.
- The paper's main legacy is the global-games resolution of equilibrium multiplicity in currency attacks.

8 Conclusion

8.1 Core conclusion

Morris and Shin show that the multiplicity of self-fulfilling currency-attack equilibria is fragile. When speculators observe slightly noisy private signals, the common-knowledge foundation of the multiplicity disappears. The result is a unique equilibrium in which both speculative attacks and government abandonment are governed by cutoffs.

8.2 Economic intuition

The intuition is that a currency crisis is a coordination problem. Each speculator wants to attack only if enough others attack. With perfect common knowledge, many collective beliefs can be self-confirming. With noisy private information, each speculator must infer not only the fundamentals but also the signals and beliefs of others. This higher-order uncertainty disciplines beliefs and produces a unique marginal signal.

8.3 Methodological contribution

The paper is an early and influential application of global-games logic to international finance. It shows that adding arbitrarily small private noise can be more than a technical perturbation: it changes the equilibrium set from many equilibria to one. This provides a microfoundation for comparative statics in crisis models.

8.4 Policy implications

The model has three main policy lessons. First, public communication can stabilize markets if it restores common knowledge. Second, transaction costs can reduce attacks, but their quantitative effect depends on the sensitivity of the exchange rate to fundamentals. Third, controls on hot money may matter most when fundamentals are opaque and speculators interpret information differently.

8.5 Interpretation for crisis timing

The paper explains why crises may occur after seemingly minor news. The news need not worsen fundamentals directly. It can instead disturb higher-order beliefs by making traders uncertain about how others interpret the situation. A rumor or ambiguous announcement can shift the coordination threshold even if average first-order beliefs remain optimistic.

8.6 Limitations

The model is static, assumes a continuum of speculators, abstracts from reserves and monetary policy dynamics, and assumes a simple signal structure. It also treats the government's payoffs as known functions. These simplifications make the model analytically transparent but limit direct empirical implementation.

8.7 Takeaway

The enduring takeaway is that transparency and common knowledge are central to financial stability. A market can be full of well-informed traders and still be fragile if traders are uncertain about what others know. Currency crises are therefore not merely about weak fundamentals or irrational panic; they are about strategic uncertainty over others' beliefs.